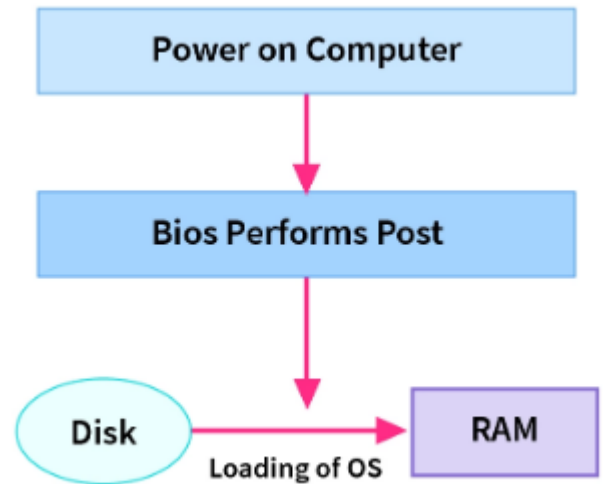


Booting an Operating System

Whenever we press the power button of our computer system, all the devices get the power and they are initialized. Our main memory which is responsible to hold the instructions will be initially empty as RAM is volatile memory. So, there will be a small set of instructions present in the non-volatile memory called ROM. These instructions will be passed to CPU and the execution of instructions takes place which will check all the hardware connected with the system. If there are any problems with the hardware, we will get the alert by beep sounds or even on-screen messages. After the testing of hardware is completed, the booting process continues and loads the operating system.

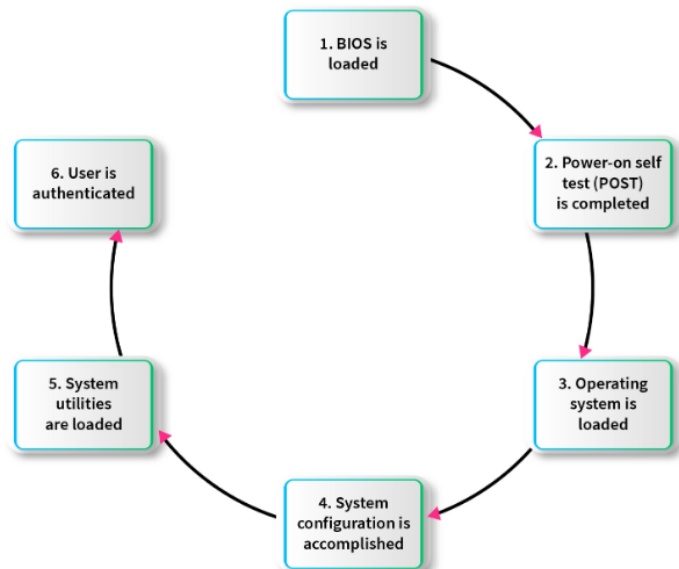


- The instructions present in non-volatile memory are hardwired on motherboard so it can't be erased.
- The small set of instructions present in the ROM is called BIOS which stands for Basic Input Output System.

Let's look into detail, the steps involved in the booting process of operating system now.

Boot Sequence in OS (Steps in Booting) :

When computer is powered on, all the hardware components receive the power and they get initialized. After that, the computer system goes through 6 step booting process as follows:

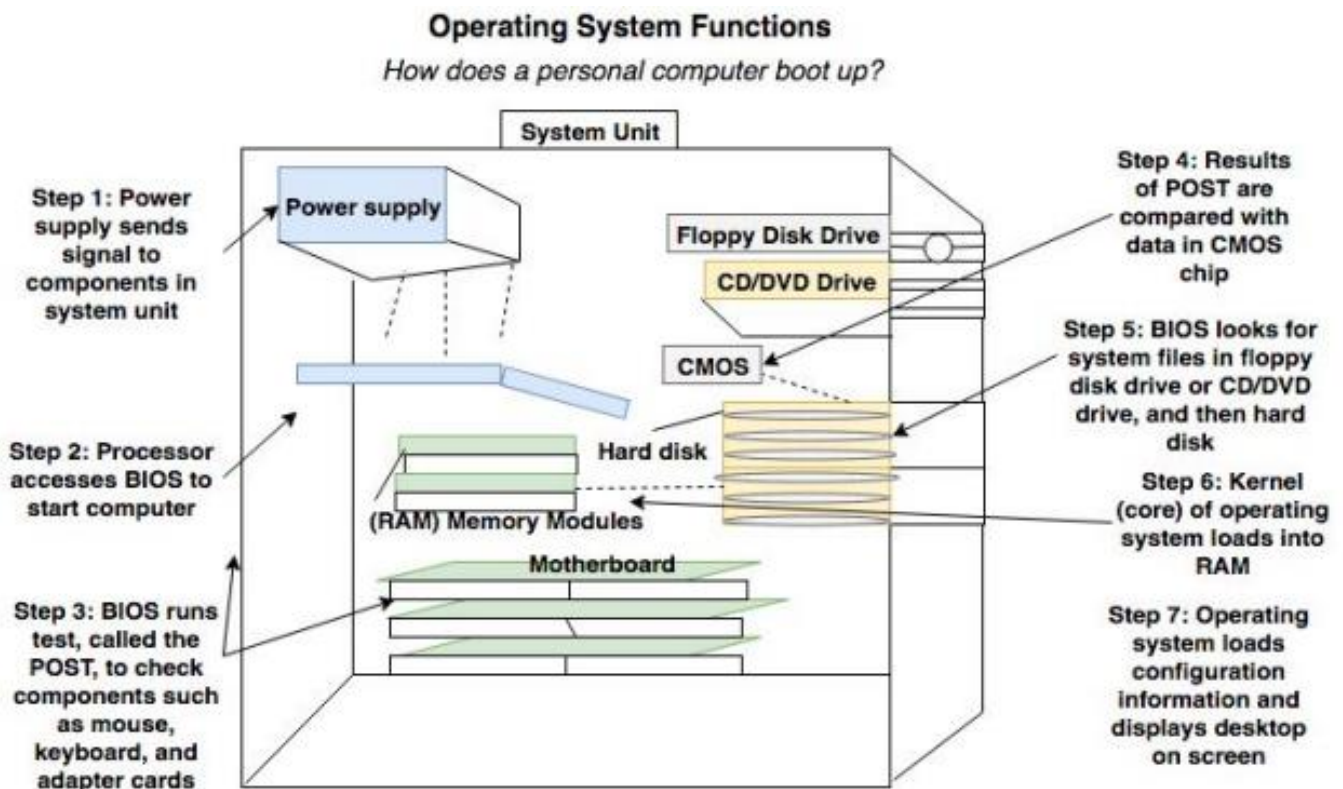


1. Loading of BIOS: The small set of instructions present in the ROM is loaded into the computer memory and CPU executes those instructions.
2. Power-On Self Test (POST): In order to check the operability of all the hardware connected to our computer system, BIOS carries out POST which will check the hardware components and if any problem is found user is alerted with POST beeps and POST screen messages.

3. Loading of Operating System:

- After the successful completion of POST, the bootable sequence present in CMOS (Common Metal Oxide Semiconductor) is read by BIOS.
 - Based on the bootable sequence it will search for Master Boot Record (MBR) in bootable device like floppy disk, CD-ROM and hard disk.
 - If MBR is not found in any of them, the system will halt by displaying “No Boot Device Found”.
 - if MBR is found, the BIOS will load the special application program called Boot Loader, that will eventually load the Operating system.
4. System Configuration is Accomplished: After the OS is loaded, device drivers are loaded into the memory so that our devices can function correctly.
 5. System Utilities are Loaded: System utilities like antivirus, volume control etc. are loaded into the memory in this step
 6. User Authentication: If any user authentication is being set, the system will ask the user to enter the credentials and on receiving the correct credentials the computer system will run GUI shell (in most cases) or CLI shell.

As BIOS is lightweight, it will just load the Boot Loader which can load the Complex set of libraries required for loading the Operating System. BIOS can't directly load the heavy weighted set of instructions responsible for loading the Operating System.



Type of Booting:

There are two types of Booting available:

1. Cold Booting/ Hard Booting:

Cold booting is the process when our computer system moves from shut down state to the start by pressing the power button. The system reads the BIOS from ROM and will eventually load the Operating System.

2. Warm Booting/ Soft Booting :

Warm booting is the process in which the computer gets restart due to reasons like setting the configuration for newly installed software or hardware. Warm booting is called as rebooting.

When our system get's hang and we restart the device, we are doing the Warm booting.

How does a System Boot?

- When we press the power button, all the components of system get power and they get initialized. Once the CPU is initialized, it needs instruction to be executed.
- The small set of instructions called BIOS are being loaded from ROM.
- After the successful completion of Power-On-Self-Test (POST) BIOS find the bootable sequence from CMOS.
- Based on the bootable sequence, it finds the first bootable device.
- From the first bootable device, it loads instructions present from Master Boot Record which is present in logical Sector 0.
- This set of instructions contains the information about the Boot Loader which can load the Operating System.
- This boot loader information is Operating System specific, for example the boot loader for Linux is GRUB (GRand Unified Bootloader).
- For apple OS BootX is used to prepare the computer for use, by loading all required device drivers and then starting-up Mac OS X by booting the kernel on all PowerPC Macintoshes running the Mac OS X 10.2 operating system or later versions.
- The boot loader then load the operating system into the memory.
- Lastly all the important system files and drivers are loaded into memory and the control is being passed to operating system.